Python Programming Lab. [Batch-B3]

Assignment Submission Details

Field	Details
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Course Name	Python Programming Lab. (MCA31PC06)
Academic Year	2025-26 (Semester-1)
Course Teacher	Prof. Prakash Ukhalkar
Assignment Name	Assignments based on modules in Python 2.
Assignment Number	Assignment 07
Submission Date *	18-11-2025

1.Program to Work with Dates (Using datetime Module)

In [1]:
```
from datetime import date

today = date.today()
print("Today's date:", today)

print("Day:", today.day)
print("Month:", today.month)
print("Year:", today.year)
```

```
Today's date: 2025-11-18
Day: 18
Month: 11
Year: 2025
```

2.Program to Generate Random Numbers (Using random Module)

In [2]:
```
import random

num = random.randint(1, 100)
print("Random number between 1 and 100:", num)
```

```
Random number between 1 and 100: 38
```

3.Program to Display Calendar of a Month (Using calendar Module) .

In [20]:
```
import calendar
```

```
year = int(input("Enter year: "))
month = int(input("Enter month (1-12): "))

print("Given year",year)
print("Given month",month)
print("-"*50)
if 1 <= month <= 12:
    print(calendar.month(year, month))
else:
    print("Please enter a value between 1 and 12.")
```

```
Given year 2025
Given month 11
--------------------------------------------------
   November 2025
Mo Tu We Th Fr Sa Su
                1  2
 3  4  5  6  7  8  9
10 11 12 13 14 15 16
17 18 19 20 21 22 23
24 25 26 27 28 29 30
```

4. Program to Calculate the Area and Circumference of a Circle.

In [7]:
```
import math

radius = float(input("Enter radius: "))
area = round(math.pi * radius * radius,2)
circumference = round(2 * math.pi * radius, 2)

print("Radius:", radius)
print("Area:", area)
print("Circumference:", circumference)
```

```
Radius: 5.0
Area: 78.54
Circumference: 31.42
```

5.Program to Generate a Random Number Between Two Numbers

In [9]:
```
import random

start = int(input("Enter start number: "))
end = int(input("Enter end number: "))

print("start number: ",start)
print("end number: ",end)

print("Random number:", random.randint(start, end))
```

```
start number:  5
end number:  100
Random number: 11
```

6.Program to Get Current Date and Time

In [11]:
```
from datetime import datetime

now = datetime.now()

date_time = now.strftime("%d-%m-%Y %H:%M:%S")

print("Current date and time:", date_time)
```

```
Current date and time: 18-11-2025 08:25:51
```

7.Program to Create a Directory and List Files in it(Using the os module)

In [23]:
```
import os

dirname = "myfolder2"


if not os.path.exists(dirname):
    os.mkdir(dirname)
    print("Directory created:", dirname)

    files = ["file1.txt", "file2.txt", "file3.txt"]

    for filename in files:
        file_path = os.path.join(dirname, filename)
        open(file_path, "w").close()
        print(f"{filename} created inside directory: {dirname}")

else:
    print("Directory already exists.")

print("Files in", dirname, ":")
print(os.listdir(dirname))
```

```
Directory created: myfolder2
file1.txt created inside directory: myfolder2
file2.txt created inside directory: myfolder2
file3.txt created inside directory: myfolder2
Files in myfolder2 :
['file1.txt', 'file2.txt', 'file3.txt']
```

8. Program to Shuffle a List of Numbers(Using the random module).

In [12]:
```
import random

numbers = [1, 2, 3, 4, 5, 6, 7]
print("original list",numbers)
random.shuffle(numbers)
print("Shuffled list:", numbers)
```

```
original list [1, 2, 3, 4, 5, 6, 7]
Shuffled list: [4, 7, 1, 5, 2, 3, 6]
```

9.Program to Display the Calendar for a Specific Month(Using the calendar module)

In [15]:
```
import calendar

year = int(input("Enter year: "))
month = int(input("Enter month (1-12): "))

print("Given year",year)
print("Given month",month)
print("-"*50)
if month < 1 or month > 12:
    print("Invalid month .enter a value between 1 and 12.")
else:
    print(calendar.month(year, month))
```

```
Given year 2025
Given month 11
--------------------------------------------------
   November 2025
Mo Tu We Th Fr Sa Su
                1  2
 3  4  5  6  7  8  9
10 11 12 13 14 15 16
17 18 19 20 21 22 23
24 25 26 27 28 29 30
```

10. Program to Perform Arithmetic Operations Using a Custom Module (Create a module math_operations.py with arithmetic functions) .

In [24]:
```
%%writefile math_umodule.py
def add(a, b):
    return a + b

def subtract(a, b):
    return a - b

def multiply(a, b):
    return a * b

def divide(a, b):
    return a / b
```

```
Overwriting math_umodule.py
```

In [ ]:
```
# we first create   math_umodule.py file in same folder and import that module

import math_umodule

a = float(input("Enter first number: "))
b = float(input("Enter second number: "))

print("First number:",a)
print("Second number:",b)


print("Addition:", math_umodule.add(a, b))
print("Subtraction:", math_umodule.subtract(a, b))
```

```python
print("Multiplication:", math_umodule.multiply(a, b))

if b != 0:
    print("Division:", math_umodule.divide(a, b))
else:
    print("Division: Cannot divide by zero")
```

```
First number: 50.0
Second number: 10.0
Addition: 60.0
Subtraction: 40.0
Multiplication: 500.0
Division: 5.0
```